



---

## **GCE A LEVEL MARKING SCHEME**

---

**A LEVEL  
CHEMISTRY  
1410U40 – 1**

---

## About this marking scheme

The purpose of this marking scheme is to provide teachers, learners, and other interested parties, with an understanding of the assessment criteria used to assess this specific assessment.

This marking scheme reflects the criteria by which this assessment was marked in a live series and was finalised following detailed discussion at an examiners' conference. A team of qualified examiners were trained specifically in the application of this marking scheme. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners. It may not be possible, or appropriate, to capture every variation that a candidate may present in their responses within this marking scheme. However, during the training conference, examiners were guided in using their professional judgement to credit alternative valid responses as instructed by the document, and through reviewing exemplar responses.

Without the benefit of participation in the examiners' conference, teachers, learners and other users, may have different views on certain matters of detail or interpretation. Therefore, it is strongly recommended that this marking scheme is used alongside other guidance, such as published exemplar materials or Guidance for Teaching. This marking scheme is final and will not be changed, unless in the event that a clear error is identified, as it reflects the criteria used to assess candidate responses during the live series.

---

**WJEC GCE A LEVEL CHEMISTRY**  
**PHYSICAL AND ORGANIC CHEMISTRY**  
**GENERAL INSTRUCTIONS**

Recording of marks

Examiners must mark in red ink.

One tick must equate to one mark (apart from the questions where a level of response mark scheme is applied).

Question totals should be written in the box at the end of the question.

Question totals should be entered onto the grid on the front cover and these should be added to give the script total for each candidate.

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statement. Award the middle mark in the level if most of the content statements are given and the communication statement is partially met. Award the lower mark if only the content statements are matched.

Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

cao = correct answer only

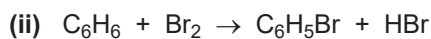
ecf = error carried forward

bod = benefit of doubt

1 (a) (i)

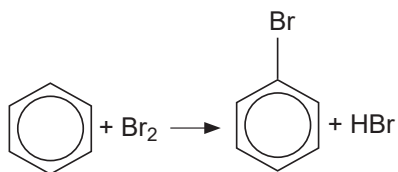
| Reaction of benzene    | Name of organic product | Name of catalyst                                 |
|------------------------|-------------------------|--------------------------------------------------|
| with bromine           | bromobenzene            | iron/iron(III) bromide/<br>aluminium bromide [1] |
| with ethanoyl chloride | phenylethanone [1]      | iron(III) chloride/<br>aluminium chloride [1]    |

[3]



[1]

or

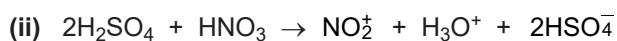


(iii) electrophilic substitution

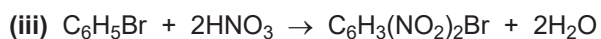
[1]

(b) (i) concentrated sulfuric acid [1]  
concentrated nitric acid [1]

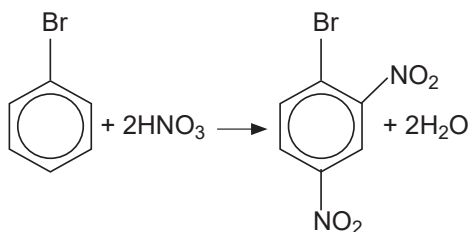
[2]



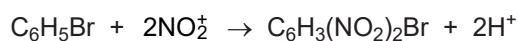
[2]



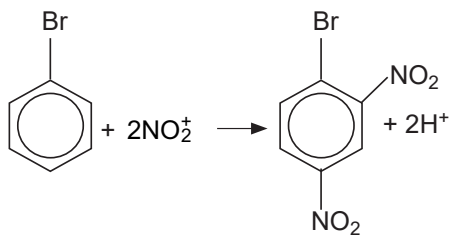
or



or



or



[1]

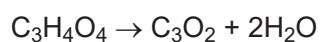
AVAILABLE  
MARKS

(a) propanedioic acid

[1]

(b) (i)  $\text{HOOCCH}_2\text{COOH} \rightarrow \text{C}_3\text{O}_2 + 2\text{H}_2\text{O}$

or



[1]

(ii)  $\text{O}=\text{C}=\text{C}=\text{C}=\text{O}$

[1]

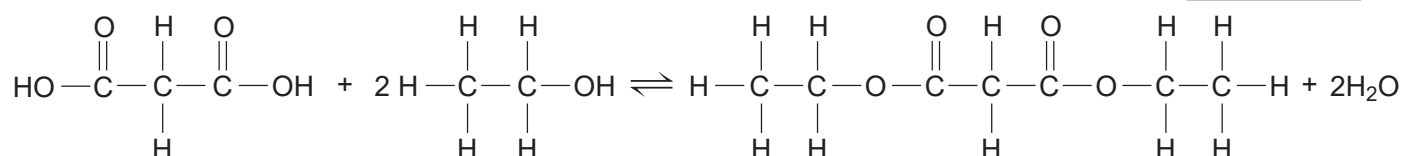
(c) (i)  $\text{HOOCCH}_2\text{COOH} + 2\text{PCl}_5 \rightarrow \text{ClOCCH}_2\text{COCl} + 2\text{POCl}_3 + 2\text{HCl}$

[2]

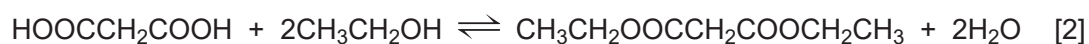
(ii) propanedioyl (di)chloride

[1]

(d) (i)



or



(ii) van der Waals' forces and permanent dipole-dipole attractions  
between molecules of ester [1]

stronger hydrogen bonds between molecules of acid [1]

[2]

(e) (i)  $\text{LiAlH}_4$

[1]

(ii) reduction

[1]

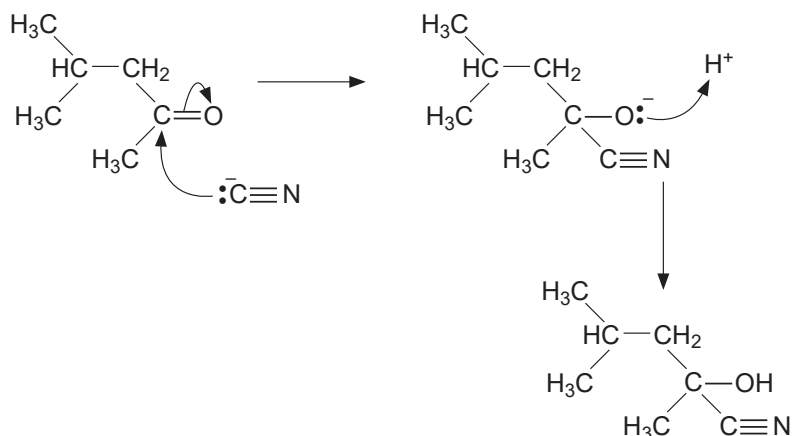
(iii)  $\text{HOOCCH}_2\text{COOH} + 8[\text{H}] \rightarrow \text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$

[2]

AVAILABLE  
MARKS

14

2 (a) (i)



3 curly arrows correct [1]

correct charges on  $\text{CN}^-$ ,  $\text{O}^-$ ,  $\text{H}^+$  [1]

lone pairs on C of  $\text{CN}^-$  and  $\text{O}^-$  of intermediate [1]

[3]

(ii) nucleophilic addition

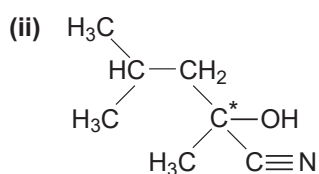
[1]

(iii) 4-methylpentan-2-one

[1]

(b) (i) an atom which has four different atoms or groups attached

[1]



[1]

(iii)  $\text{CN}^-$  attack equally probable from top or bottom/racemic mixture/equal concentrations of both optical isomers formed

[1]

(c) (i) 2-hydroxy-2,4-dimethylpentanoic acid

[2]

(ii) ammonia

[1]

(iii) ethanal

[1]

Section B

12

100

Total

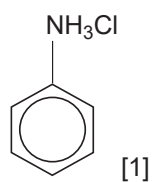
110

(b) (i) nitrobenzene [1]

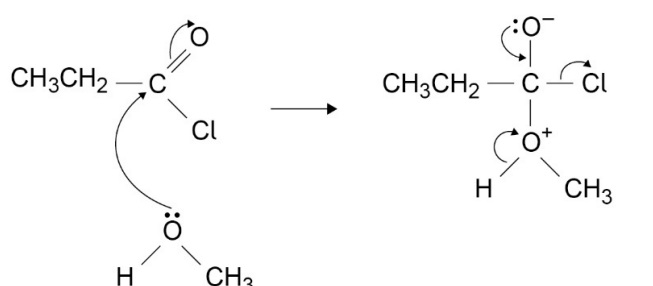
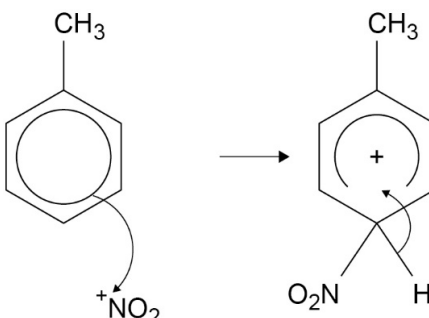
(ii) tin [1]  
concentrated hydrochloric acid [1] [2]

(iii) reduction [1]

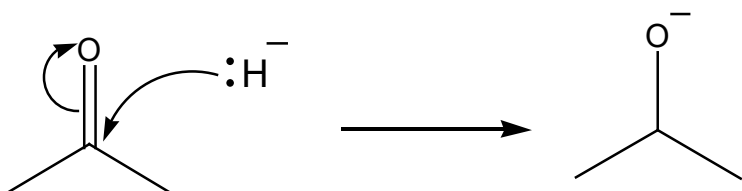
(iv)



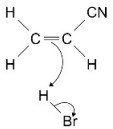
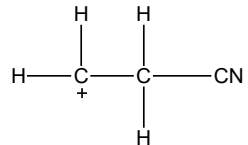
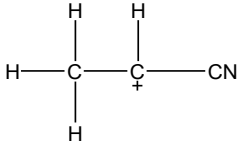
name: phenylammonium chloride [1] [2]

| Question | Answers                                                                                                                 | Additional Comments/Guidelines                                                                                                                                                                                                                                                                                                                      | Mark           |
|----------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 4 .1     |                                       | M1 Structures of reactant species<br>M2 Arrow from lp on O to C<br>M3 Arrow from C=O bond to O<br>M4 3 curly arrows and lp on intermediate<br>Allow full marks for candidates who draw a second intermediate formed after formation of C=O and loss of Cl <sup>-</sup> then loss of H <sup>+</sup><br>Ignore any attempt to show the final products | 4<br>(4 x AO1) |
| Question | Answers                                                                                                                 | Additional Comments/Guidelines                                                                                                                                                                                                                                                                                                                      | Mark           |
| 4 .2     |  <p>M4 Electrophilic substitution</p> | M1 Structures of reactant species including + on N of <sup>+</sup> NO <sub>2</sub><br>M2 Arrow from ring (inside hexagon) to N or + on N<br>M3 Arrow from C-H bond into hexagon<br>Apply list principle                                                                                                                                             | 4<br>(4 x AO1) |



| Question | Answers                                                                                                            | Additional Comments/Guidelines                                                                                                                                                                                                                                              | Mark           |
|----------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 4.3      |  <p>M4 Nucleophilic addition</p> | <p>M1 Arrow from lp on hydride to C</p> <p>M2 Arrow from C=O to O</p> <p>M3 Intermediate structure / Allow displayed or abbreviated structures<br/>Ignore any attempt to show further steps if correct<br/>Penalise further incorrect steps</p> <p>Apply list principle</p> | 4<br>(4 x AO1) |

| Question | Answers               | Additional Comments/Guidelines | Mark       |
|----------|-----------------------|--------------------------------|------------|
| 5.1      | 3-bromopropanenitrile | Allow 3-bromopropane-1-nitrile | 1<br>(AO1) |

| Question | Answers                                                                                                             |                                                                                                                                                                                                                                                                                         | Additional Comments/Guidelines                                                                                                                              | Mark                       |
|----------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 5.2      | This question is marked using Levels of Response. Refer to the Mark Scheme Instructions for Examiners for guidance. |                                                                                                                                                                                                                                                                                         | <b>Indicative Chemistry content</b>                                                                                                                         | 6<br>(3 x AO1,<br>3 x AO3) |
|          | Level 3<br>5–6 marks                                                                                                | All stages are covered and each stage is generally correct and virtually complete.<br><br>Answer is communicated coherently and shows a logical progression from Stage 1 to Stages 2 and 3.                                                                                             | <b>Stage 1 Types of Isomers formed</b><br><br>1a $\text{CH}_3\text{CHBrCN}$<br><br>1b Exists as two Optical isomers / enantiomers                           |                            |
|          | Level 2<br>3–4 marks                                                                                                | All stages are covered but stage(s) may be incomplete or may contain inaccuracies<br><b>OR</b> two stages are covered and are generally correct and virtually complete.<br><br>Answer is communicated mainly coherently and shows a logical progression from Stage 1 to Stages 2 and 3. | <b>Stage 2 Mechanism</b><br><br>2a 2 curly arrows<br><br>                |                            |
|          | Level 1<br>1–2 marks                                                                                                | Two stages are covered but stage(s) may be incomplete or may contain inaccuracies <b>OR</b> only one stage is covered but is generally correct and virtually complete.<br><br>Answer includes isolated statements but these are not presented in a logical order.                       | 2b Intermediate structure primary carbocation OR<br><br>                 |                            |
|          | 0 mark                                                                                                              | Insufficient correct chemistry to gain a mark.                                                                                                                                                                                                                                          | 2c Alternative Intermediate structure secondary carbocation OR<br><br> |                            |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  | <p><b>Stage 3 Optical isomerism</b></p> <p>3a 2-bromo isomer has chiral carbon / C with four different groups / non superimposable mirror images</p> <p>OR</p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <math display="block">\begin{array}{c} \text{CH}_3 \\   \\ \text{C} \cdots \text{CN} \\   \quad \diagup \\ \text{H} \quad \text{Br} \end{array}</math> </div> <div style="text-align: center;"> <math display="block">\begin{array}{c} \text{CH}_3 \\   \\ \text{NC} \cdots \text{C} \\ \diagdown \quad   \\ \text{Br} \quad \text{H} \end{array}</math> </div> </div> <p>3b Optical because (secondary) C<sup>+</sup> planar</p> <p>3c So can be attacked from above or below</p> <div style="text-align: center;"> </div> |  |
|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

| Question     | Answers                                            | Additional Comments/Guidelines | Mark           |
|--------------|----------------------------------------------------|--------------------------------|----------------|
| <b>5</b> 1.3 | M1 KCN or NaCN<br>M2 Aqueous AND ethanol (alcohol) | Penalise acid in M1            | 2<br>(2 x AO1) |

| Question     | Answers                                                                                                                                                       | Additional Comments/Guidelines                                                                                             | Mark                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>5</b> 1.4 | M1 H <sub>2</sub> and Ni/Pt/Pd<br>M2 NCCH <sub>2</sub> CH <sub>2</sub> CN + 4H <sub>2</sub> → H <sub>2</sub> N(CH <sub>2</sub> ) <sub>4</sub> NH <sub>2</sub> | Allow LiAlH <sub>4</sub> and (Dry) ether BUT <u>not</u> NaBH <sub>4</sub><br>(ignore heat and pressure)<br>Allow with 8[H] | 2<br>(1 x AO1,<br>1 x AO2) |

| Question | Answers                            | Additional Comments/Guidelines                                     | Mark       |
|----------|------------------------------------|--------------------------------------------------------------------|------------|
| 06.1     | Use H <sub>2</sub> SO <sub>4</sub> | Allow HCl / H <sub>3</sub> PO <sub>4</sub><br>Ignore conc / dilute | 1<br>(AO1) |

| Question | Answers                                                                                                        | Additional Comments/Guidelines                      | Mark           |
|----------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------|
| 06.2     | M1 Cool test 2<br>warm (water bath)<br>M2 Gas is tested with lighted splint in test 3<br>Bubble into limewater | Allow heat / hot<br><br>Allow no test on gas needed | 2<br>(2 x AO3) |

| Question | Answers                                                                                                                                                                                                                                                | Additional Comments/Guidelines                                                                                                                                   | Mark           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 06.3     | M1 J and M<br>M2 Test 1 (Orange solution goes) green<br>M3 M<br>M4 Test 2 (Blue solution gives a brick) red precipitate<br>M5 J and L<br>M6 Test 3 (Colourless gas that turns) limewater cloudy<br>M7 K<br>M8 Test 4 (Orange solution goes) colourless | Allow (Brown-red/orange/orange-red)<br><br>Allow M6 Test 3 fizz / effervescence<br><br>Allow (Brown/Brown-red/yellow/yellow-orange)<br>Allow decolorises bromine | 8<br>(8 x AO3) |

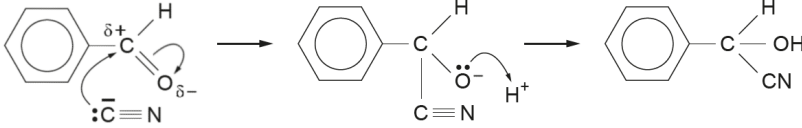
| Question       | Answers        | Additional Comments/Guidelines                          | Mark       |
|----------------|----------------|---------------------------------------------------------|------------|
| 7 <sub>1</sub> | $C_nH_{2n-2}O$ | Allow $C_nH_{2n}CO$ or $(CH_2)_nCO$ or $C_nH_{2(n-1)}O$ | 1<br>(AO2) |

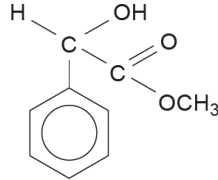
| Question       | Answers                                                                                                                               | Additional Comments/Guidelines       | Mark           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------|
| 7 <sub>2</sub> | <p>M1 curly arrow from bond to O</p> <p>M2 lone pair and curly arrow from lp to H</p> <p>M3 curly arrow from C-H bond to C-C bond</p> | Allow other C-O bond breaking for M1 | 3<br>(3 x AO2) |

| Question | Marking details | Marks available |     |     |       |       |      |
|----------|-----------------|-----------------|-----|-----|-------|-------|------|
|          |                 | AO1             | AO2 | AO3 | Total | Maths | Prac |

|   |     |       |  |                                                                                                        |   |   |   |    |   |   |
|---|-----|-------|--|--------------------------------------------------------------------------------------------------------|---|---|---|----|---|---|
| 8 | (c) | (i)   |  | LiAlH <sub>4</sub>                                                                                     | 1 |   |   | 1  |   |   |
|   |     | (ii)  |  |                       | 1 |   |   | 1  |   |   |
|   |     | (iii) |  | dehydration                  accept elimination                                                        |   |   | 1 | 1  |   |   |
|   | (d) |       |  | add NaHCO <sub>3</sub> / Na <sub>2</sub> CO <sub>3</sub> (1)<br>effervescence with the most acidic (1) |   | 2 |   | 2  |   | 2 |
|   |     |       |  | Question 7 total                                                                                       | 5 | 6 | 4 | 15 | 1 | 5 |



| Question |     |     |    | Marking details                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|-----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |     |    |                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) | (i) | I  |  <p>award (1) each for up to <b>three</b> of following</p> <p>curly arrows and addition of <math>\text{CN}^-</math></p> <p>capture of <math>\text{H}^+</math></p> <p>partial and full charges</p> <p>dissociation of <math>\text{HCN}</math> into <math>\text{H}^+</math> and <math>\text{CN}^-</math></p> |                 | 3   |     | 3     |       |      |
|          |     |     | II | nucleophilic addition                                                                                                                                                                                                                                                                                                                                                                        | 1               |     |     | 1     |       |      |

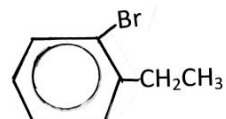
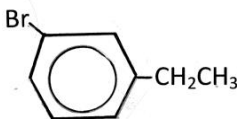
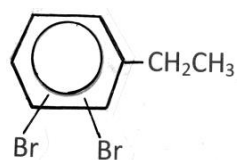
|  |  |       |    |                                                                                                                     |  |   |   |   |  |  |
|--|--|-------|----|---------------------------------------------------------------------------------------------------------------------|--|---|---|---|--|--|
|  |  | (iii) |    | it is a racemic mixture / an equimolar mixture of the two enantiomers                                               |  |   | 1 | 1 |  |  |
|  |  | (iv)  |    | it does not contain a chiral centre / no asymmetric carbon atom                                                     |  | 1 |   | 1 |  |  |
|  |  | (v)   | I  | it acts as a carboxylic acid<br> |  | 1 |   | 2 |  |  |
|  |  |       | II | it acts as an alcohol (giving an ester)                                                                             |  | 1 |   | 1 |  |  |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) | (i)  |  | orange to green (accept colourless to green)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"> <li>• distillate placed in a separating funnel</li> <li>• add water to the distillate</li> <li>• add ethoxyethane to the funnel</li> <li>• equal volume of ethoxyethane as distillate</li> <li>• (stopper) and shake</li> <li>• allow to separate</li> <li>• run off lower aqueous layer</li> <li>• treat ethoxyethane layer with anhydrous magnesium sulfate</li> <li>• remove magnesium sulfate (by filtration/decanting)</li> <li>• remove ethoxyethane by distillation</li> <li>• use of heating mantle</li> </ul> <p><b>5-6 marks</b><br/>All the essential steps described in logical order to ensure a dry sample of pentan-2-one<br/><i>The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i></p> <p><b>3-4 marks</b><br/>Most of the correct steps described (although order may not be correct); safe method involving no naked flames<br/><i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i></p> <p><b>1-2 marks</b><br/>Some correct steps in separation method<br/><i>The candidate attempts to link at least two relevant points from the indicative material. Coherence is limited by omission and/or inclusion of irrelevant materials. There is some evidence of appropriate use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give an answer worthy of credit.</i></p> |                 | 3   | 3   | 6     |       | 6    |

| Question |  |       |  | Marking details                                                                                                                                                                | Marks available |           |          |           |          |          |
|----------|--|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|----------|-----------|----------|----------|
|          |  |       |  |                                                                                                                                                                                | AO1             | AO2       | AO3      | Total     | Maths    | Prac     |
|          |  | (iii) |  | no peak seen at 3200-3500 cm <sup>-1</sup> (due to the OH group in the alcohol)<br><br>accept no peak seen at 1000-1300 cm <sup>-1</sup> (due to the C—O group in the alcohol) |                 | 1         |          | 1         |          |          |
|          |  | (iv)  |  | alkaline iodine / KI and NaOCl (1)<br><br>yellow precipitate / solid forms (1)<br><br>do <b>not</b> accept 2,4-DNPH test                                                       |                 | 2         |          | 2         |          | 2        |
|          |  |       |  | <b>Question 10 total</b>                                                                                                                                                       | <b>3</b>        | <b>12</b> | <b>5</b> | <b>20</b> | <b>0</b> | <b>9</b> |

|    | Question |      |  | Marking details                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----|----------|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|    |          |      |  |                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10 | (a)      | (i)  |  | (deep) blue Fehling's solution (1)<br>gives a red / brown precipitate (1)                                                                                                                                                                                                                                                     | 2               |     |     | 2     |       | 2    |
|    |          | (ii) |  | award (1) for name/formula of any aliphatic aldehyde e.g.<br>propanal / CH <sub>3</sub> CH <sub>2</sub> CHO<br>do not accept benzaldehyde / C <sub>6</sub> H <sub>5</sub> CHO                                                                                                                                                 |                 | 1   |     | 1     |       |      |
|    | (b)      | (i)  |  | nucleophilic addition                                                                                                                                                                                                                                                                                                         | 1               |     |     | 1     |       |      |
|    |          | (ii) |  | decomposition by water / dilute acid                                                                                                                                                                                                                                                                                          | 1               |     |     | 1     |       |      |
|    | (c)      |      |  | award (1) each for any <b>two</b> of following<br>loss of (volatile) ethanal<br>incomplete oxidation / further oxidation<br>side reactions                                                                                                                                                                                    |                 |     | 2   | 2     |       | 2    |
|    | (d)      |      |  | moles of NaOH = $0.200 \times \frac{40.0}{1000} = 0.0080 \text{ mol}$ (1)<br>moles of lactic acid reacted = 0.0080 mol<br>mass of lactic acid reacted = $0.0080 \times 90.06 = 0.72 \text{ g}$ (1)<br>volume of lactic acid = $\frac{0.72}{1.2} = 0.60 \text{ cm}^3$<br>percentage = $\frac{0.60}{75} \times 100 = 0.8\%$ (1) |                 | 2   | 1   | 3     | 1     |      |



| Question |     |       |    | Marking details                                                                                                                                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|-----|-------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |    |                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 12       | (a) | (i)   |    | bromine and iron(III) bromide / Br <sub>2</sub> and FeBr <sub>3</sub>                                                                                                                                                                                                            |                 | 1   |     | 1     |       | 1    |
|          |     | (ii)  |    | 17                                                                                                                                                                                                                                                                               | 1               |     |     | 1     |       | 1    |
|          |     | (iii) | I  | award (1) for both of following structures<br><div></div><br>substitution can occur elsewhere in the ring (1) | 1               | 1   |     | 2     |       |      |
|          |     |       | II |                                                                                                                                                                                                 |                 | 1   |     | 1     |       |      |
|          | (b) |       |    | alcoholic sodium hydroxide solution (1)<br>a small molecule / HBr is removed (1)                                                                                                                                                                                                 | 1               | 1   |     | 2     |       | 2    |
|          | (c) |       |    | add aqueous bromine dropwise (1)<br>bromine decolourised (1)                                                                                                                                                                                                                     | 1               | 1   |     | 2     |       | 2    |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (d) | (i)   |  | to react with benzoic acid (1)<br><br>to give water soluble sodium benzoate (1)<br><br>(other possible correct answers to discussed at the examiners' conference)                                                                              |                 |     | 2   | 2     |       |      |
|          |     | (ii)  |  | the density of trichloromethane is greater than the density of the aqueous layer                                                                                                                                                               |                 | 1   |     | 1     |       | 1    |
|          |     | (iii) |  | moles of phenylethene = $\frac{3.00}{104} = 0.289 \text{ mol}$ (1)<br><br>moles of phenyloxirane = $\frac{2.50}{120} = 0.208 \text{ mol}$ (1)<br><br>mole ratio is 1:1<br><br>percentage yield = $\frac{0.208}{0.289} \times 100 = 71.9\%$ (1) |                 | 3   |     | 3     | 2     |      |
|          | (e) |       |  | a peak would be seen between 1650 and 1750 $\text{cm}^{-1}$ (1)<br><br>indicating the presence of the C=O group (1)                                                                                                                            | 2               |     |     | 2     |       |      |

| Question                                                                                                      |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     | Marks available                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|---------------------------------------------------------------------------------------------------------------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|-----------------|------------------------|---------------------------|------------------|----------------------|------------------------|---------------------------------------------|-------------------------------------|------------------------|--|---|---|--|---|
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     | AO1                                                                 | AO2             | AO3                    | Total                     | Maths            | Prac                 |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               | (f) |  |  | <table><tr><td>Test</td><td>phenylethanal<br/>C<sub>6</sub>H<sub>5</sub>CH<sub>2</sub>CHO</td><td>phenylethanone<br/>C<sub>6</sub>H<sub>5</sub>C(O)CH<sub>3</sub></td></tr><tr><td>alkaline iodine</td><td>no observable reaction</td><td>yellow precipitate formed</td></tr><tr><td>Tollens' reagent</td><td>silver mirror formed</td><td>no observable reaction</td></tr><tr><td>warming with acidified potassium dichromate</td><td>solution turns from orange to green</td><td>no observable reaction</td></tr></table> | Test                                                               | phenylethanal<br>C <sub>6</sub> H <sub>5</sub> CH <sub>2</sub> CHO  | phenylethanone<br>C <sub>6</sub> H <sub>5</sub> C(O)CH <sub>3</sub> | alkaline iodine | no observable reaction | yellow precipitate formed | Tollens' reagent | silver mirror formed | no observable reaction | warming with acidified potassium dichromate | solution turns from orange to green | no observable reaction |  | 3 | 3 |  | 3 |
|                                                                                                               |     |  |  | Test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | phenylethanal<br>C <sub>6</sub> H <sub>5</sub> CH <sub>2</sub> CHO | phenylethanone<br>C <sub>6</sub> H <sub>5</sub> C(O)CH <sub>3</sub> |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  | alkaline iodine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | no observable reaction                                             | yellow precipitate formed                                           |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  | Tollens' reagent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | silver mirror formed                                               | no observable reaction                                              |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  | warming with acidified potassium dichromate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | solution turns from orange to green                                | no observable reaction                                              |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
| award (3) for all correct<br>award (2) for any four or five correct<br>award (1) for any two or three correct |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                    |                                                                     |                                                                     |                 |                        |                           |                  |                      |                        |                                             |                                     |                        |  |   |   |  |   |
|                                                                                                               |     |  |  | Question 8 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                    |                                                                     | 6                                                                   | 12              | 2                      | 20                        | 2                | 10                   |                        |                                             |                                     |                        |  |   |   |  |   |

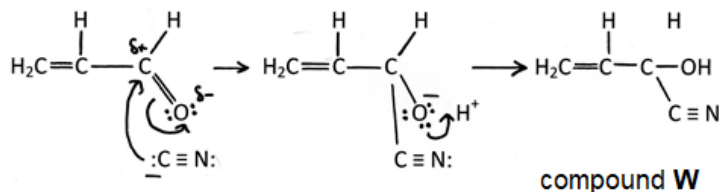


| Question | Marking details | Marks available |     |     |       |       |      |
|----------|-----------------|-----------------|-----|-----|-------|-------|------|
|          |                 | AO1             | AO2 | AO3 | Total | Maths | Prac |

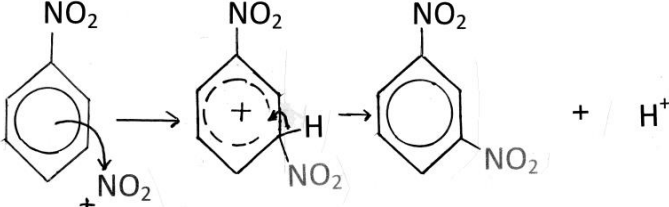
|    |     |  |  |                                                                                                                                                                                                                                          |   |   |   |    |   |   |
|----|-----|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|----|---|---|
| 13 | (f) |  |  | $  \begin{array}{ccccc}  & \text{H} & & \text{H} & & \text{CH}_3 \\  &   & &   & &   \\  \text{HO} & - \text{C} & - & \text{C} & - & \text{C} & - \text{OH} \\  &   & &   & &   \\  & \text{H} & & \text{H} & & \text{H}  \end{array}  $ |   |   | 1 | 1  |   |   |
|    |     |  |  | Question 9 total                                                                                                                                                                                                                         | 2 | 7 | 9 | 18 | 1 | 1 |

| Question |     |     |     | Marking details                                                                                                                                | Marks available |     |     |       |       |      |
|----------|-----|-----|-----|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |     |     |                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 14       | (b) | (i) | I   | $\text{ClCH}_2(\text{CH}_2)_2\text{CH}_2\text{Cl} + 2\text{KCN} \rightarrow \text{NC}(\text{CH}_2)_4\text{CN} + 2\text{KCl}$                   |                 | 1   |     | 1     |       |      |
|          |     |     | II  | $\text{CN}^-$                                                                                                                                  | 1               |     |     | 1     |       |      |
|          |     |     | III | award (1) for any of following<br><br>lithium tetrahydridoaluminate(III) / $\text{LiAlH}_4$<br>hydrogen with Ni catalyst<br>sodium and ethanol | 1               |     |     | 1     |       | 1    |

| Question | Marking details | Marks available |     |     |       |       |      |
|----------|-----------------|-----------------|-----|-----|-------|-------|------|
|          |                 | AO1             | AO2 | AO3 | Total | Maths | Prac |

|    |  |     |    |                                                                                                                                                             |   |   |  |   |  |  |
|----|--|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|---|--|--|
| 15 |  | (v) | I  |  <p>award (1) for appropriate charges<br/>award (1) for curly arrows</p> |   | 2 |  | 2 |  |  |
|    |  |     | II | <p>initial attack is by the nucleophile <math>\text{CN}^-</math> (1)</p> <p>HCN adds across the C=O bond (1)</p>                                            | 2 |   |  | 2 |  |  |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                     | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                     | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) | (i)   |  | lithium tetrahydridoaluminate(III) / LiAlH <sub>4</sub> (1)<br>hydrogen is added / oxygen is lost (1)                                                                                                                                                               | 2               |     |     | 2     |       |      |
|          |     | (ii)  |  | benzene-1,4-dioic acid<br>accept terephthalic acid                                                                                                                                                                                                                  |                 | 1   |     | 1     |       |      |
|          | (c) | (i)   |  | $  \begin{array}{c} \text{COO}^-\text{Na}^+ \\   \\ \text{CH}_2 \\   \\ \text{COO}^-\text{Na}^+ \end{array} + 2\text{NaOH} \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{CH}_4  $                                                                                 |                 | 2   |     | 2     |       |      |
|          |     | (ii)  |  | sodium hydrogencarbonate / NaHCO <sub>3</sub> (1)<br>accept sodium carbonate / Na <sub>2</sub> CO <sub>3</sub><br><br>reaction complete when no more effervescence seen (1)<br><br>phenol does not react with sodium hydrogencarbonate (to give carbon dioxide) (1) |                 | 3   |     | 3     |       | 3    |
|          |     | (iii) |  | no purple coloration seen                                                                                                                                                                                                                                           |                 |     | 1   | 1     |       | 1    |
|          |     |       |  | Question 11 total                                                                                                                                                                                                                                                   | 5               | 12  | 1   | 18    | 1     | 4    |

| Question |     |       |    | Marking details                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|-------|----|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |    |                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 16       | (b) | (i)   |    |                                                 |                 | 2   |     | 2     |       |      |
|          |     | (ii)  | I  | Fe / Sn and hydrochloric acid                                                                                                     | 1               |     |     | 1     |       | 1    |
|          |     |       | II | NaOH / sodium hydroxide                                                                                                           | 1               |     |     | 1     |       | 1    |
|          |     | (iii) |    | HNO <sub>2</sub> / NaNO <sub>2</sub> , HCl /<br>nitric(III) acid / sodium nitrate(III) and hydrochloric acid<br><br>0-10°C        | 2               |     |     | 2     |       | 2    |
|          |     | (iv)  |    | amine group in the azo dye has a lone pair of electrons on the nitrogen atom and acts as an electron pair donor / proton acceptor |                 |     | 1   | 1     |       |      |

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (IGC HK Exam) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

IGC HK Exam – WJEC is part of the IGC HK, which is itself a department of IGC HK.